1. Corpus, Research Question, Literature Review

Yelp-reviews is one of the largest crowd-sourced business review platforms. With the increasing divide between Apple and Google centered platforms, Yelp is the default and got-to review hub on Apple devices. These reviews offer a great look into textual analysis, sentiment, and ratings. Yelp reviews also feature a voting system in which users could upvote a review as “funny”, “cool”, “usefull”, or any combination of the three.

Finding a relationship between different aspects can give insights on designing review functions on other forms of online reviews. For my research question, I am investigating if Yelp reviews that are more negative correlate with being perceived as more useful? Some research has found that very positive or very negative reviews are considered more useful than moderate reviews (Roh 2019). When looking through a binary sentiment lens, negative reviews were considered more useful than positive ones. Other research on sentiment and usefulness have found that useful negative reviews are more likely to be connected to conversations on the cost and value for restaurant experiences (Luo 2019).

2. Hand Coding, Inter-coder reliability

Because I am interested in negative reviews, my binary coding for reviews if 1 if they are more negative and 0 if they are more positive. My codebook has the following guidelines:

Overall
- Sentences at the beginning and end of a paragraph often represent user opinion more strongly than those in the middle.
- For reviews that feel neutral/mid, code according to whatever binary sentiment is emphasized with more details or greater intensity

Negative = 1
- Even if they say the place is usually good but describe a uniquely poor experience due to something momentary like a bad server or parking, mark as negative

Positive = 0
- the concluding sentence carries the most weight as it represents overall opinion. If you see mixed sentiment but it ends with something like "would go again", mark as positive
- If neutral and talking more about stuff like weird parking/service/location quirks, “round up” to positive seeing as how they did not provide a strong/substantial complaint

The inter-coder reliability was quite satisfactory with a Krippendorff’s alpha of 0.875. Coder 1 marked 28% of the randomly selected one-hundred reviews as negative, and coder 2 marked

27%. Because I was coder 1, I will consider my coding as “true” and the standard for comparison. The precision rate was 89.3% meaning the hand coding of negative reviews was quite strong when looking at all negative guesses. The recall rate was 92.6%.This further emphasizes satisfactory inter-coder reliability.

There were a total of 5 reviews where codes conflicted. There were coding inconsistencies because the coding guideline did not explicitly state whether the proportion of positive-negative sentiment throughout the full review weighed over conclusive ending sentences or not. The other type of coding conflict came from humorous or light hearted complaints. Reviewers would sometimes mention downsides that did not personally affect themselves but would have dampened the experience for others. Bringing up what would be a con for others but not weighing it into personal rating was not considered in my initial coding guidelines.

Based on these two types of conflicts, I would revise my coding book to note that proportion of negative-to-positive sentiment expressed in the reviews should outweigh conclusion sentences or rating. Regardless of the reviewers’ personal conclusion, negative details and criticisms are still shared for other’s usage. In other words, negative sentiment regardless of final ratings are written with the goal to inform other users to reach their own conclusions. This ties in better to my research question if negative reviews are more informative and useful. Again, weighing the text’s proportion of sentiment expressed weighs heavier even with conflicting humor in reviews. This approach best adapts sentiment in relation to usefulness.

Step 2: Measurement [Option A]

I trained my classifier using naive bayes. I chose this model over lasso regression because of my cross validation within the hand coding sample. I tested four models, naive bayes, lasso with a lambda of 60, ten-fold lasso, and upweighted negative reviews on ten-fold lasso.

When looking at out of sample precision and recall, my naive bayes model had the highest F1 score of 0.65 and the highest recall rate of 0.83. Though a simple lasso regression with a higher lambda of 60 resulted in the highest precision rate of 0.60, I decided to care more about the F1 score as it takes into account both precision and recall.

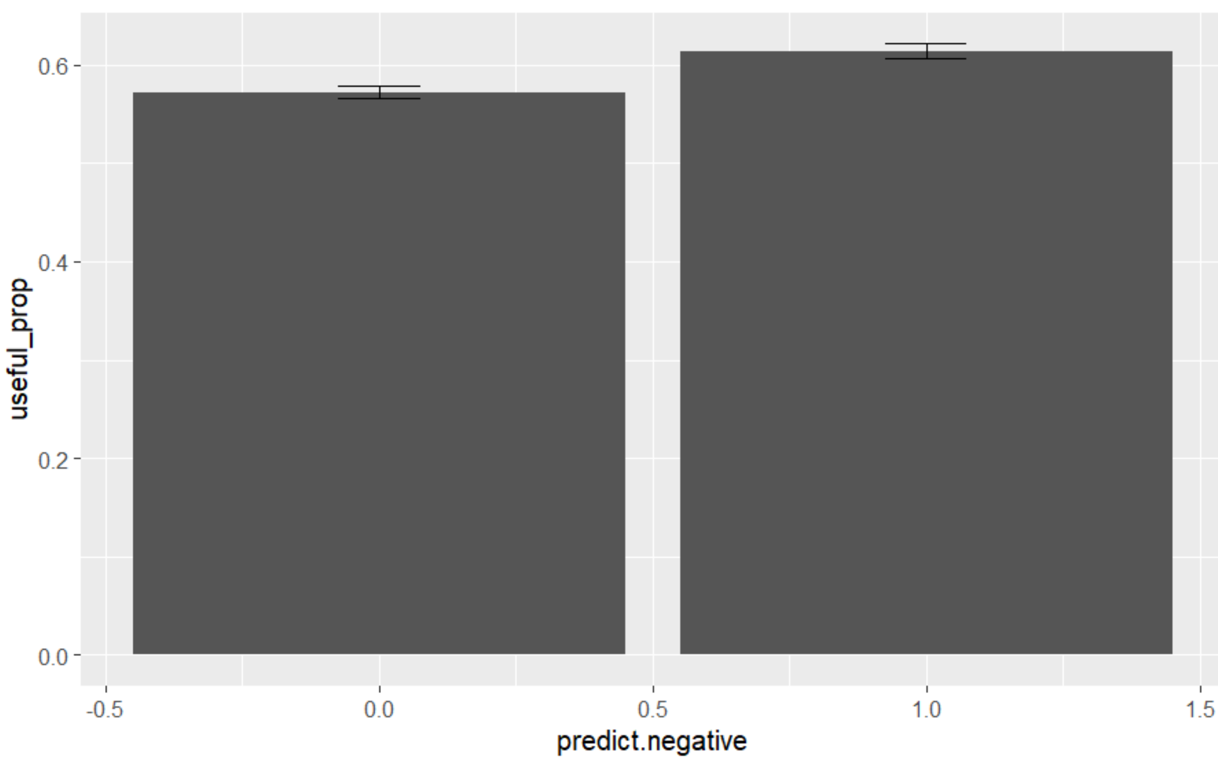
Step 3: Apply Option A to the Whole Corpus

I applied my naive bayes model to the entirety of my 10,000 yelp review dataset. This resulted in 36% of the corpus predicted as more negative. Compared to my overall hand coded sample of 30% negative reviews, my model over-predicted by about 6%.

In order to explore possible relationships between more negative reviews and useful votes, I had to alter the useful column data. The useful column ranged 0 to 76 with a median of 1 and a mean of 1.409. This means that the useful variable is incredibly right-skewed. As such, I logged that variable to make it less right-skewed. I fit a linear regression model to predict log_useful from the negative prediction scores. The model was horrendous with a p-value of 0.855, and an adjusted R-squared of 0.0004. This model shows that we fail to reject the null hypothesis that there is no relationship between log_useful and predicted pos/neg reviews.

I then made a binary version of the logged useful numbers. I set it so that above average logged usefulness would be recorded at 1 and below average at 0. With my predicted negative categorization and useful variable both binary, I looked at the proportion of above to below average logged useful scores grouped by positive and negative reviews. The results showed that 57% of above average useful reviews were positive and 61% of below average useful reviews were negative. Looking at the proportion of above-below useful reviews by binary sentiment classification did not answer much for my research question.

The figure below highlights that negative reviews(x=1) have a higher proportion of above average useful reviews. This suggests that negative reviews are perceived as somewhat more useful than positive reviews.



Some limitations of this approach is that making continuous values into binary oversimplifies the data and it loses out on many linguistic nuances. Additionally natural language processing will always struggle with context, sarcasm, and other forms of fluid and informal language. These issues could be improved by using LLM classification as it can consider context to a greater degree than naive bayes or lasso could ever. Trying other predictive algorithms such as random forest and trees holds the possibility of better predictive precision and recall rates.

Works Cited

Roh, M., & Yang, S.-B. (2021). Exploring extremity and negativity biases in online reviews: Evidence from Yelp.com. Social Behavior and Personality: An International Journal. https://doi.org/10.2224/sbp.10825

Luo, Y., & Xu, X. (2019). Predicting the Helpfulness of Online Restaurant Reviews Using Different Machine Learning Algorithms: A Case Study of Yelp. Sustainability, 11(19), 5254. https://doi.org/10.3390/su11195254